Light

GCSE Computer Science — Data Representation: Images and Sound (Set 2)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What type of image is made up of a grid of pixels?

- ○ **A.** A bitmap image
- ○ **B.** A vector image
- ○ **C.** An audio file
- ○ **D.** A text document

Q2 (1 mark)

What type of signal is a sound wave before it is recorded digitally?

- ○ **A.** Binary

- B. Hexadecimal
- C. Analogue
- D. Compressed

Q3 (1 mark)

If a pixel uses 4 bits to store its colour, how many different colours can that pixel display?

- A. 4
- B. 8
- C. 32
- D. 16

Q4 (1 mark)

During sound sampling, what property of the sound wave is measured?

- A. The amplitude (height) of the wave
- B. The colour of the wave
- C. The temperature of the wave
- D. The weight of the wave

Q5 (1 mark)

An image has a resolution of 640 × 480. What does this tell us?

- ○ **A.** The image has 640 pixels wide and 480 pixels tall
- ○ **B.** The image uses 640 colours and 480 shades of grey
- ○ **C.** The file size is 640 KB and takes 480 seconds to load
- ○ **D.** The image has 640 bits of colour and 480 bits of data

Q6 (1 mark)

What unit is used to measure sample rate?

- ○ **A.** Bytes per second
- ○ **B.** Bits per pixel
- ○ **C.** Pixels per inch
- ○ **D.** Hertz (Hz)

Q7 (1 mark)

If an image's resolution is doubled (both width and height), what happens to the number of pixels?

- ○ **A.** It doubles
- ○ **B.** It triples
- ○ **C.** It stays the same
- ○ **D.** It quadruples (×4)

Q8 (1 mark)

What does increasing the bit depth of a sound recording improve?

- ○ **A.** The length of the recording
- ○ **B.** The number of samples per second
- ○ **C.** The precision of each amplitude measurement
- ○ **D.** The speed of playback

Q9 (1 mark)

In a 24-bit colour image, how many bits are typically used for each colour channel (Red, Green, Blue)?

- ○ **A.** 4 bits each
- ○ **B.** 6 bits each
- ○ **C.** 8 bits each
- ○ **D.** 24 bits each

Q10 (1 mark)

Why must sound be converted to binary for a computer to store it?

- ○ **A.** Binary makes sound louder
- ○ **B.** Sound waves are already in binary format
- ○ **C.** Binary compresses the sound automatically

- ○ **D.** Computers can only process and store data in binary format (0s and 1s)

Q11 (1 mark)

An artist notices visible banding (distinct colour steps) in a gradient on their digital image. Which setting should they increase to reduce this?

- ○ **A.** Sample rate
- ○ **B.** Colour depth
- ○ **C.** File name length
- ○ **D.** Screen brightness

Q12 (1 mark)

A recording sounds muffled and is missing high-pitched sounds. Which setting was likely too low during recording?

- ○ **A.** Volume level
- ○ **B.** Sample rate
- ○ **C.** File size
- ○ **D.** Playback speed

Q13 (2 marks)

Calculate the file size in bytes for an image with resolution 1024 × 768 pixels and 8-bit colour depth.

- ○ **A.** 786,432 bytes
- ○ **B.** 1,572,864 bytes
- ○ **C.** 6,291,456 bytes
- ○ **D.** 98,304 bytes

Q14 (2 marks)

A mono audio recording uses a sample rate of 22,050 Hz and 8-bit depth. Calculate the file size for a 5-second recording.

- ○ **A.** 13,781 bytes
- ○ **B.** 220,500 bytes
- ○ **C.** 882,000 bytes
- ○ **D.** 110,250 bytes

Q15 (2 marks)

A digital photograph is reduced from 3000 × 2000 pixels to 1500 × 1000 pixels. Describe the effect on file size and image detail.

- ○ **A.** File size halves; detail is unchanged
- ○ **B.** File size quarters; some fine detail is lost
- ○ **C.** File size doubles; detail improves
- ○ **D.** File size quarters; detail improves

Q16 (2 marks)

Explain why increasing the sample rate from 22,050 Hz to 44,100 Hz improves audio quality.

- ○ **A.** It makes each sample louder
- ○ **B.** It increases the bit depth automatically
- ○ **C.** It reduces the file size by half
- ○ **D.** It doubles the number of samples per second, capturing more detail of the sound wave's changes and allowing higher frequencies to be accurately recorded

Q17 (2 marks)

An image uses 16-bit colour depth. How many different colours can be displayed, and how does this compare to 8-bit colour?

- ○ **A.** 256 colours; double the amount of 8-bit
- ○ **B.** 65,536 colours; 256 times more than 8-bit
- ○ **C.** 16 colours; half of 8-bit
- ○ **D.** 4,096 colours; 16 times more than 8-bit

Q18 (2 marks)

A stereo audio file is recorded instead of mono. What effect does this have on the file size?

- ○ **A.** No effect — stereo and mono are the same size
- ○ **B.** File size doubles because two channels of audio data are stored

- ○ **C.** File size halves because stereo is more efficient
- ○ **D.** File size increases by 50%

Q19 (2 marks)

A web developer saves a logo image as a PNG file. What type of compression does PNG use, and why is this suitable for logos?

- ○ **A.** Lossy compression; suitable because logos need small file sizes
- ○ **B.** No compression; PNG files are always uncompressed
- ○ **C.** Lossy compression; suitable because logos have complex colour gradients
- ○ **D.** Lossless compression; suitable because no image data is lost, preserving sharp edges and solid colours

Q20 (2 marks)

A recording is changed from 16-bit to 24-bit depth. By what factor does the storage per sample increase?

- ○ **A.** 1.5 times (50% increase)
- ○ **B.** 2 times (100% increase)
- ○ **C.** 8 times
- ○ **D.** 256 times

Q21 (2 marks)

A photograph's metadata shows it was taken with GPS coordinates embedded. What concern might this raise?

- ○ **A.** The image quality will be reduced
- ○ **B.** The file cannot be opened on other devices
- ○ **C.** Privacy concern — the location where the photo was taken is revealed when the image is shared
- ○ **D.** The image will load more slowly

Q22 (2 marks)

According to Nyquist's theorem, what is the maximum frequency that can be accurately captured when sampling at 48,000 Hz?

- ○ **A.** 48,000 Hz
- ○ **B.** 96,000 Hz
- ○ **C.** 12,000 Hz
- ○ **D.** 24,000 Hz

Q23 (2 marks)

Image A is 500 × 400 pixels with 24-bit colour. Image B is 1000 × 800 pixels with 8-bit colour. Which has the larger file size?

- ○ **A.** Image A is larger
- ○ **B.** Image B is larger
- ○ **C.** They are exactly the same size

○ **D.** Cannot be determined without more information

Q24 (3 marks)

A security camera system records footage 24 hours a day. Each frame is 1920 × 1080 pixels with 24-bit colour, captured at 25 frames per second.
Analyse the storage challenge this presents for one hour of uncompressed footage.

○ **A.** Approximately 175 MB per hour — easily manageable with a small USB drive

○ **B.** Approximately 17.5 GB per hour — significant storage requirement needing regular maintenance or cloud storage

○ **C.** Approximately 527 GB per hour — requiring massive storage infrastructure, which is why security systems use heavy compression

○ **D.** Approximately 1.75 GB per hour — standard hard drive would store months of footage

Q25 (3 marks)

A podcast producer records a 1-hour interview in uncompressed WAV format (44,100 Hz, 16-bit, stereo). The client needs an MP3 version for distribution.
Analyse the typical file sizes and trade-offs involved.

○ **A.** WAV: approximately 635 MB; MP3: approximately 55-90 MB at standard quality; trade-off is permanent loss of audio detail for much smaller distribution size

○ **B.** WAV: approximately 63 MB; MP3: approximately 6 MB; no quality difference between formats

○ **C.** WAV and MP3 would be the same size since they contain the

same audio

○ **D.** MP3 would be larger than WAV because it contains additional metadata

Q26 (3 marks)

A photographer must choose between saving images as RAW (30 MB each, 14-bit colour, no compression) or JPEG (5 MB each, 8-bit colour, lossy compression).
Analyse the trade-offs for professional photography work.

○ **A.** JPEG is always better because smaller files are easier to manage

○ **B.** RAW is always better because larger files mean better photos

○ **C.** RAW provides maximum editing flexibility with 16,384 brightness levels per channel and no compression artefacts, but requires 6× more storage and post-processing; JPEG is convenient for immediate sharing but limits editing potential due to reduced colour depth and permanent compression losses

○ **D.** There is no real difference — both formats produce identical final images

Q27 (3 marks)

A music producer notices that quiet sections of an 8-bit recording have an audible 'grainy' texture not present in the original performance.
Analyse what causes this and how it could be prevented.

○ **A.** The recording equipment was faulty and needs replacement

○ **B.** This is quantisation noise — with only 256 amplitude levels, the rounding errors between the actual amplitude and the nearest available level create audible distortion, especially noticeable in quiet passages where the signal-to-noise ratio is low; using 16-bit or 24-bit

depth would provide thousands more levels, reducing rounding errors to inaudibility

○ **C.** The sample rate was too high, capturing unwanted room noise

○ **D.** The microphone picked up electromagnetic interference from nearby electronics

Q28 (3 marks)

A graphic designer creates an illustration in vector format. The client requests both a 300 DPI print version and a small web version.
Analyse how vector format helps meet both requirements efficiently.

○ **A.** Vector format cannot produce print-quality images, so a new bitmap must be created from scratch

○ **B.** The vector file must be duplicated — one copy for print and one for web

○ **C.** The single vector file can be exported at any resolution without quality loss: scaled up for high-DPI print output and scaled down for web; because vectors store mathematical shape descriptions rather than pixels, the rendering engine calculates appropriate pixels at export time for any size

○ **D.** Vector files only work for web, not print

Q29 (3 marks)

A music streaming service offers 'lossless' streaming at 1,411 kbps alongside standard streaming at 256 kbps AAC.
Analyse the practical differences for typical listeners using smartphone speakers or standard earbuds.

○ **A.** Lossless always sounds dramatically better regardless of playback equipment

○ **B.** Lossless streaming uses approximately 5.5× the data of standard AAC; however, for typical listeners on smartphone speakers or standard earbuds, the quality difference is often imperceptible because the playback hardware cannot reproduce the subtle details preserved by lossless; high-quality headphones or speakers are needed to benefit from lossless

○ **C.** Standard streaming actually sounds better because compression removes unwanted noise

○ **D.** There is no difference — both formats contain identical audio data

Q30 (3 marks)

Both digital images and digital sound involve converting continuous analogue information into discrete digital values.
Analyse the parallel concepts between image digitisation (resolution and colour depth) and sound digitisation (sample rate and bit depth).

○ **A.** There are no similarities — image and sound digitisation are completely different processes

○ **B.** Resolution corresponds to bit depth; colour depth corresponds to sample rate

○ **C.** Resolution (how many samples in space) parallels sample rate (how many samples in time) — both determine how finely the continuous source is divided; colour depth (precision per pixel) parallels bit depth (precision per sample) — both determine how accurately each discrete measurement represents the original continuous value

○ **D.** Image digitisation is analogue; sound digitisation is digital

Total: **55** marks • Time: **30 min**

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